

Addendum A10 — Stereopsis of the Dual Cross-Slit

Companion to “Double-Slit Geometry: Definitions, Axioms, and Theorems” and to Addendum A8 / demo 34. All numerical claims verified by `check_stereopsis.js` (node, no dependencies; 22 assertions, all passing). Rig: shared screen $z = 0$ and shared z -axis; channel A has $\ell_1 = \{(0, y, 150)\}$ and $\ell_2 = \{(x, 0, 250)\}$; channel B is channel A rotated 90° about the shared axis, $\ell_1' = \{(x, 0, 150)\}$ and $\ell_2' = \{(0, y, 250)\}$. $\Delta = Z_h - Z_v = 100$.

Figure A10-1: the dual cross-slit construction (K.M., pencil sketch, 2026-07-04): cross slit + cross slit at 90° = double cross-slit projection, with the tetrahedron, the neutral surface, and the squares at infinity marked. The sketch is the figure for this addendum and for the second tetrahedron of the atlas (§9 candidate).

A10.0 Claim

The dual cross-slit supports depth perception analogous to stereoscopic vision, and in two mechanically distinct ways at once: a **positional channel**, in which the two rigs restore pairwise the concurrence that a single rig refuses (Axiom C2), yielding true triangulation with an epipolar consistency constraint; and a **shape channel**, in which disparity lives in aspect ratio rather than position, with the neutral surface playing the role of the horopter. Both channels obey the same depth-acuity law, and that law is quantitatively the $b \cdot f / z^2$ law of binocular stereo, with the slit separation Δ serving as an *axial* baseline.

A10.1 The positional channel: triangulation restored

Within one rig no two rays meet (C2). Across the pair, concurrence returns pointwise: a scene point P off both seams is threaded by exactly one ray of each channel, and the two rays intersect precisely at P .

Verified. Over 200 random scene points, the two channel rays concur at the point with worst gap 1.8×10^{-14} , are distinct lines (genuine triangulation), and degenerate exactly on the blind locus: for P on the shared axis the two rays coincide and triangulation fails, as the two “eyes” share a sightline.

The epipolar identity. The pair of channels measures four numbers (s_A, t_A, s_B, t_B) for a point with three degrees of freedom. The single consistency constraint is

$$s_A \cdot t_A = s_B \cdot t_B,$$

verified to relative residual 3.5×10^{-16} on random points. The constraint is a product identity — the two channels must agree on the *product* of a point's address coordinates, a form with the same algebraic flavor as the split norm of The Seam §5.

Two further consequences, both verified:

1. **A positional depth coordinate.** The cross-channel ratios agree and depend on depth alone: $\rho = s_A/t_B = s_B/t_A = -(z - Z_v)/(z - Z_h)$, independent of (X, Y) to 10^{-12} . ρ is a Möbius function of z — and it is the aperture cross-ratio of Theorem 7 with the screen point traded for the point at infinity: $(F_1, F_2; \infty, P) = -1/\rho$. The positional channel reads depth at Level 2 (affine) where Theorem 7 reads it at Level 3 (calibrated). No target extent is required: ρ is defined for a mathematical point, which the shape channel's AR is not.
2. **The gap is the residual.** For corrupted measurements, the 3D gap between the two reconstructed rays is first-order in the measurement error (log-log slope 1.0000) and proportional to the epipolar residual $|s_A t_A - s_B t_B|$ with a stable configuration-dependent constant (3.76×10^{-2} at the test configuration). The epipolar identity is therefore a usable rejection test: mismatched correspondences announce themselves as ray gap, exactly as in classical stereo.

The positional channel inherits stereo's correspondence problem (which blob in channel B is this blob in channel A); the combined cross-aperture's channel-arm assignment (A8) solves it by construction for the shape channel but not for point features, which must be matched. The epipolar identity is the matching test.

A10.2 The shape channel: reciprocal aspect and the horopter

The 90° rotation swaps the roles of the two magnifications, so the channels measure exactly reciprocal aspect ratios:

$$AR_B(z) = 1 / AR_A(z),$$

verified by direct projection of an infinitesimal square through both channels (worst $|AR_A \cdot AR_B - 1| = 1.5 \times 10^{-10}$; channel A reproduces Theorem 7's $AR(350) = 0.3$ to nine decimals). The disparity signal is the ratio

$$D(z) = AR_A/AR_B = AR_A(z)^2,$$

and the zero-disparity surface — the locus where the two channels fuse, $D = 1$ — is found by blind root-search at

$$z^* = 2 \cdot Z_v \cdot Z_h / (Z_v + Z_h) = 187.500000000,$$

the neutral surface: the harmonic-conjugate locus ($F_1, F_2; S, P$) = -1 (verified). The geometry's harmonic surface is the rig's **shape-horopter**, playing the role fixation plays in binocular vision — with the honest difference that it is fixed by the rig; vergence would mean physically moving slit depths.

A10.3 The stereoacuity law

Differentiating the disparity signal gives the addendum's central formula. The log-disparity depth sensitivity is

$$S(z) = d(\log D)/dz = 2\Delta / ((z - Z_v)(z - Z_h)),$$

verified against numerical differentiation to relative error 2.2×10^{-9} across the working range. Three verified readings of it:

1. **At the horopter**, $|S(187.5)| = 200/(37.5 \times 62.5) \approx 0.0853$: about 8.5% disparity change per unit depth. Acuity is highest where the channels fuse — as in binocular vision, where stereoacuity peaks at the fixation surface.
 2. **In the far field**, $S(z) \cdot z^2 \rightarrow 2\Delta$ (measured 200.80 at $z = 10^5$ against the limit 200). This is the $b \cdot f/z^2$ law of stereo vision: depth acuity falling as inverse depth squared, with constant set by the baseline. The dual cross-slit's baseline is the slit separation Δ — a baseline pointing *down the optical axis* rather than across it. The analogy with two eyes is not qualitative; it is the same asymptotic law with a rotated baseline.
 3. **The two channels compared**. The positional coordinate obeys $d(\log \rho)/dz = -S(z)/2$ exactly: the shape channel is precisely twice as depth-sensitive as the positional channel on a log scale, everywhere. Both share the $(z - Z_v)(z - Z_h)$ denominator: acuity diverges at the slit planes (where the codec is in any case invalid — the divergence is formal, not operational) and both decay together in the far field ($|d\rho/dz| \cdot z^2 \rightarrow \Delta$, measured 100.50).
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A10.4 What the second channel buys, and the dictionary

Confound cancellation (re-verified in this configuration). A target ellipticity of 23% compounded with a 7% PSF asymmetry multiplies both channels' aspect measurements identically, so it cancels exactly in D : the single channel is fooled by 92.44 units at $z = 350$, while the dual ratio recovers z with error below 10^{-9} . (Demo 34's corresponding impostor figure, 162.75 units, was measured at its own configuration; the cancellation, not the

magnitude, is the invariant statement.)

$\sqrt{2}$. With independent per-channel noise, averaging the two channel estimates gains $\sqrt{2}$ in depth SNR (measured 1.434 on 8000 deterministic-PRNG trials, uniform relative noise 10^{-3}), consistent with demo 34's 1.414.

Binocular vision	Dual cross-slit
two eyes	two channels, 90° apart about the shared axis
positional disparity	shape disparity $D = AR^2$ (and positional p)
horopter (fusion locus)	neutral surface $z^* = 187.5$, the harmonic locus
epipolar constraint	$s_A \cdot t_A = s_B \cdot t_B$; violation = ray gap
correspondence problem	solved by channel-arm assignment (shape); epipolar test (points)
stereoacuity $\propto b \cdot f / z^2$	$S(z) \cdot z^2 \rightarrow 2\Delta$; baseline = slit separation, axial
stereoacuity peaks at fixation	$ S $ maximal at the horopter
disparity noise $\rightarrow \sigma_z = \sigma / S $	same, with the closed-form $S(z)$ above

Honest disanalogies. The shape channel requires extended targets — a mathematical point has no aspect ratio — trading stereo's correspondence problem for the roundness prior (the positional channel covers points, at half the log-acuity). The horopter is fixed; there is no vergence. And unlike an eye, a single channel already measures absolute depth (Theorem 7): the second channel buys robustness — cancellation, the epipolar test, $\sqrt{2}$ — rather than creating depth from nothing. The construction is closer to how binocular vision hardens monocular cues than to how it conjures depth without them.

A10.5 Deferred

The D-solver's own bias analysis ($\sqrt{}$ -concavity of $AR = \sqrt{D}$ against Möbius convexity of $z(AR)$) remains with A8's open items, together with off-axis orientation and chromatic mismatch tolerance. The present addendum fixes the frame those analyses will live in.

Verification summary

Claim	Result
channel rays concur at scene points; distinct lines	worst gap 1.8×10^{-14}
blind locus on shared axis (rays coincide)	verified
epipolar identity $s_A \cdot t_A = s_B \cdot t_B$	rel. residual 3.5×10^{-16}
$\rho = s_A/t_B = s_B/t_A = -(z-Z_v)/(z-Z_h)$, (X,Y)-free	to 10^{-12} ; $\rho(337) = -2.149425287$
$\rho = -1/(F_1, F_2; \infty, P)$	verified
gap first-order in error; $\propto$ epipolar residual	slope 1.0000; const 3.76×10^{-2}
$AR_B = 1/AR_A$ by direct projection	worst 1.5×10^{-10}
horopter $D = 1$ at $z^* = 187.5$ = harmonic locus	187.500000000
$S(z) = 2\Delta/((z-Z_v)(z-Z_h))$	rel. err 2.2×10^{-9}
far-field $S \cdot z^2 \rightarrow 2\Delta$; positional $\rightarrow \Delta$	200.80; 100.50
shape channel = $2 \times$ positional (log acuity)	exact, verified
impostor 23%/7%: single fooled, dual exact	92.44 units vs $< 10^{-9}$
$\sqrt{2}$ SNR gain from channel averaging	1.434 measured

`check_stereopsis.js` — 22 assertions, 0 failures.